

Enumeration of linear codes with different hulls

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WCC, Paris, June 2026

- 1 **Linear codes**
- 2 **Inner products and hulls**
- 3 **Enumeration of linear codes with different hulls**
 - Euclidean hulls
 - Hermitian hulls
 - Symplectic hulls

- Hulls are fundamental invariants of linear codes that depend on the underlying inner product.
- Understanding the distribution of hull dimensions helps reveal the global structure of code families.
- Previous work established enumeration results for codes with prescribed Euclidean hulls.
- We extend these investigations to Hermitian and symplectic hulls.
- Using mass formulae, we derive counting results and inequalities analogous to those known in the Euclidean setting, leading to a unified treatment of the three hull notions.

- $\mathbb{F}_q$ - finite field with q elements;
- C is a linear $[n, k, d]_q$ code if it is a k -dimensional subspace of $\mathbb{F}_q^n$ with $d(C) = \min\{\text{wt}(x) \mid x \in C, x \neq \mathbf{0}\}$
- $(u, v) : \mathbb{F}_q^n \times \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ be an inner product in the linear space $\mathbb{F}_q^n$;
- The dual code of C is $C^\perp = \{u \mid (u, v) = 0 \ \forall v \in C\}$
- C is a *self-orthogonal* code, if $C \subseteq C^\perp$
If C is self-orthogonal, then $k \leq n/2$
- C is a *self-dual* code, if $C = C^\perp$
If C is self-dual, then $k = n/2$
- C is a *Linear Complementary Dual (LCD)* code, if $C \cap C^\perp = \{0\}$
If C is an LCD code, then $C^\perp$ is also LCD

- *Euclidean (standard) inner product* (symmetric bilinear form):

$$x \cdot y = \sum_{i=1}^n x_i y_i,$$

for any $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n) \in \mathbb{F}_q^n$.

- *Hermitian inner product* (sesquilinear form):

$$\langle x, y \rangle_H = \sum_{i=1}^n x_i y_i^q, \quad x, y \in \mathbb{F}_{q^2}^n.$$

Hermitian product is defined only over fields with q^2 elements.

- *Symplectic inner product* (alternating bilinear form):

$$\langle x, y \rangle_S = \sum_{i=1}^n (x_i y_{n+i} - x_{n+i} y_i), \quad x, y \in \mathbb{F}_q^{2n}.$$

Symplectic product is defined only in vector spaces of even dimension.

The hull of a linear code

The hull of a linear code C is $\mathcal{H}(C) = C \cap C^\perp$

$$\Rightarrow 0 \leq \dim \mathcal{H}(C) \leq \min\{k, n - k\}$$

$\dim \mathcal{H}(C) = 0 \iff C$ is LCD, $\dim \mathcal{H}(C) = k \iff C$ is self-orthogonal

We consider $\mathcal{H}_E(C)$, $\mathcal{H}_H(C)$, and $\mathcal{H}_S(C)$.

Example

Let C be a linear $[6, 2]$ code over $\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}$, where $1 + \omega + \omega^2 = 0$, generated by

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & \omega & \omega^2 & 1 \end{pmatrix}.$$

C is Euclidean self-orthogonal $\Rightarrow \mathcal{H}_E(C) = C \Rightarrow \dim \mathcal{H}_E(C) = 2$,
 $\mathcal{H}_H(C) = \langle (1 \ 0 \ 1 \ 1 \ 1 \ 0) \rangle \Rightarrow \dim \mathcal{H}_H(C) = 1$,
 $\mathcal{H}_S(C) = \{0\} \Rightarrow C$ is symplectic LCD code.

- $EHULL[n, k, \ell]_q$ - the number of all different linear $[n, k]_q$ codes with Euclidean hull of dimension ℓ ($\leq \ell \leq k \leq n/2$).
- $HHULL[n, k, \ell]_{q^2}$ - the number of all different linear $[n, k]$ codes over $\mathbb{F}_{q^2}$ with Hermitian hull of dimension ℓ ($\leq \ell \leq k \leq n/2$).
- $SHULL[2n, k, \ell]_q$ - the number of all different linear $[2n, k]_q$ codes with symplectic hull of dimension ℓ ($\leq \ell \leq k \leq n$). If $k - \ell$ is odd then $SHULL[2n, k, \ell]_q = 0$.

Enumeration of linear codes with different hulls

Consider the sequences

$$\{EHULL[n, k, 0]_q, EHULL[n, k, 1]_q, \dots, EHULL[n, k, k]_q\}$$

$$\{HHULL[n, k, 0]_{q^2}, HHULL[n, k, 1]_{q^2}, \dots, HHULL[n, k, k]_{q^2}\}$$

$$\{SHULL[2n, k, \underbrace{k - 2\lfloor k/2 \rfloor}_{\ell_1}]_q, SHULL[2n, k, \ell_1 + 2]_q, \dots, SHULL[2n, k, k]_q\}$$

We prove that for fixed n, k and q , and for all $0 \leq \ell \leq k - 1$:

- 1 All three sequences are strongly decreasing.
- 2 We find rationals μ_E, μ_H and μ_S such that

$$\begin{aligned} EHULL[n, k, \ell]_q &\geq \mu_E[n, k, \ell]_q EHULL[n, k, \ell + 1]_q \\ HHULL[n, k, \ell]_{q^2} &\geq \mu_H[n, k, \ell]_{q^2} EHULL[n, k, \ell + 1]_{q^2} \\ SHULL[2n, k, \underbrace{k - 2\ell_0}_{\ell}]_q &\geq \mu_S[n, k, \ell]_q SHULL[2n, k, \ell + 2]_q \end{aligned}$$

$\sigma_q(n, k)$ - the number of all different q -ary self-orthogonal codes of length n and dimension k

Theorem (Vera Pless 1965)

Let $m = \lfloor n/2 \rfloor$ and $\pi_{n,k} = \prod_{i=1}^k \frac{q^{2m-2i+2}-1}{q^i-1}$. For all $1 \leq k \leq m$, we have

$$\sigma_q(n, k) = \begin{cases} \pi_{n,k}, & \text{if } n \text{ is odd,} \\ \frac{q^{n-k}-1}{q^n-1} \pi_{n,k}, & \text{if } n \text{ and } q \text{ are even,} \\ \frac{q^{m-k}-1}{q^m-1} \pi_{n,k}, & \text{if } n \equiv 2 \pmod{4} \text{ and } q \equiv 3 \pmod{4}, \\ \frac{q^{m-k}+1}{q^m+1} \pi_{n,k}, & \text{if } (n \equiv 0 \pmod{4}) \\ & \text{or } (n \equiv 2 \pmod{4} \text{ and } q \equiv 1 \pmod{4}). \end{cases}$$

$$q = 2 \Rightarrow \sigma_2(2m, m) = (2^{m-1} + 1)(2^{m-2} + 1) \cdots (2 + 1)$$

$$q = 2 \Rightarrow \sigma_2(6, 3) = 15$$

$\sigma_q(n, k)$ - the number of all different q -ary self-orthogonal codes of length n and dimension k

Theorem (mass formula)

For all $1 \leq k \leq n/2$, we have

$$\sigma_2(n, k) = \sum_{C \in \mathcal{B}_2(n, k)} \frac{n!}{|\text{Aut}(C)|},$$

where $\mathcal{B}_2(n, k)$ is the set of all binary self-orthogonal $[n, k]$ codes.

Example: There is only one self-orthogonal $[6, 3]$ binary code C :

$$\text{gen}(C) = \begin{pmatrix} 110000 \\ 001100 \\ 000011 \end{pmatrix}, \quad |\text{Aut}(C)| = 48 \Rightarrow \sigma_2(6, 3) = \frac{6!}{48} = 15$$

$T_2(n, k)$ - the number of all different binary LCD codes of length n and dimension k

Theorem (Carlet et al. 2019)

For all $1 \leq k \leq n-1$, we have

$$T_2(n, k) = \begin{cases} 2^{(nk-k^2+n-1)/2} \begin{bmatrix} n/2-1 \\ (k-1)/2 \end{bmatrix}_4, & \text{if } n \text{ is even and } k \text{ is odd,} \\ 2^{(n-k)(k+1)/2} \begin{bmatrix} (n-1)/2 \\ (k-1)/2 \end{bmatrix}_4, & \text{if } n \text{ and } k \text{ are odd,} \\ 2^{k(n-k+1)/2} \begin{bmatrix} (n-1)/2 \\ k/2 \end{bmatrix}_4, & \text{if } n \text{ is odd and } k \text{ is even,} \\ 2^{k(n-k)/2} \left(2^{n-k} \begin{bmatrix} n/2-1 \\ k/2-1 \end{bmatrix}_4 + \begin{bmatrix} n/2-1 \\ k/2 \end{bmatrix}_4 \right), & \text{if } n \text{ and } k \text{ are even,} \end{cases}$$

where $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is the Gaussian binomial coefficient, defined by

$$\begin{bmatrix} n \\ 0 \end{bmatrix}_q = 1, \quad \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^{n-1} - 1) \dots (q^{n-k+1} - 1)}{(q^k - 1)(q^{k-1} - 1) \dots (q - 1)}, \text{ if } 1 \leq k \leq n.$$

$T_3(n, k)$ - the number of all different ternary LCD codes of length n and dimension k

Theorem (Carlet et al. 2019)

For all $1 \leq k \leq n - 1$, we have

$$T_3(n, k) = \begin{cases} 3^{(nk-k^2+n-3)/2} \begin{bmatrix} n/2-1 \\ (k-1)/2 \end{bmatrix}_9, & \text{if } n \equiv 0 \pmod{4} \text{ and } k \text{ is odd,} \\ 3^{(nk-k^2+n+1)/2} \begin{bmatrix} n/2-1 \\ (k-1)/2 \end{bmatrix}_9, & \text{if } n \equiv 2 \pmod{4} \text{ and } k \text{ is odd,} \\ 3^{(n-k)(k+1)/2} \begin{bmatrix} (n-1)/2 \\ (k-1)/2 \end{bmatrix}_9, & \text{if } n \text{ and } k \text{ are odd,} \\ 3^{k(n-k+1)/2} \begin{bmatrix} (n-1)/2 \\ k/2 \end{bmatrix}_9, & \text{if } n \text{ is odd and } k \text{ is even,} \\ 3^{k(n-k)/2} \begin{bmatrix} n/2 \\ k/2 \end{bmatrix}_9, & \text{if } n \text{ and } k \text{ are even.} \end{cases}$$

$EHULL[n, k, \ell]_q$ - the number of all different q -ary linear $[n, k]$ codes whose Euclidean hull has dimension ℓ , $0 \leq \ell \leq k$.

Theorem (Sendrier 1997)

Let k and $n \geq 2k$ be positive integers, and $0 \leq \ell \leq k$. Then

$$EHULL[n, k, \ell]_q = \sum_{i=\ell}^k \begin{bmatrix} n-2i \\ k-i \end{bmatrix} \begin{bmatrix} i \\ \ell \end{bmatrix} (-1)^{i-\ell} q^{\binom{i-\ell}{2}} \sigma_q(n, i). \quad (1)$$

$$EHULL[n, k, k]_q = \sigma_q(n, k)$$

$$EHULL[n, k, 0]_2 = T_2(n, k), \quad EHULL[n, k, 0]_3 = T_3(n, k)$$

The average dimension of the hull

$EHULL[n, k, \ell]_q$ - the number of all different q -ary linear $[n, k]$ codes whose hull has dimension ℓ , $0 \leq \ell \leq k$.

Theorem (Sendrier 1997)

For all ℓ , the proportion $EHULL[n, k, \ell]_q / \begin{bmatrix} n \\ k \end{bmatrix}_q$ is convergent when $n \rightarrow \infty$ and $k \rightarrow \infty$. If R_ℓ is the limit, then

$$R_\ell = \frac{R_0}{(q-1)(q^2-1)\dots(q^\ell-1)} = \frac{R_{\ell-1}}{q^\ell-1}, \quad R_0 = \prod_{i=1}^{\infty} \frac{q^i}{q^i+1}. \quad (2)$$

Theorem (Sendrier 1997)

The average dimension of the hull of a q -ary linear code is asymptotically equal to

$$\sum_{\ell \geq 1} \ell R_\ell = \sum_{i \geq 1} \frac{1}{q^i+1}. \quad (3)$$

The average dimension of the hull (Sendrier 1997)

The average dimension of the hull of a q -ary linear code is asymptotically equal to

- $q = 2$) 0.7645
- $q = 3$) 0.404063
- $q = 4$) 0.2794
- $q = 5$) 0.215062

In all cases the average dimension of the hull is $< 1!!!$

Hence most of the linear $[n, k]_q$ codes are LCD!!!

Theorem (Carlet, Mesnager, Tang, Qi and Pellikaan)

Any code over $\mathbb{F}_q$ for $q \geq 4$ is equivalent to an Euclidean LCD code.

Any code over $\mathbb{F}_{q^2}$ for $q \geq 3$ is equivalent to an Hermitian LCD code.

$EHULL[n, k, \ell]_q$ - the number of all different q -ary linear $[n, k]$ codes whose hull has dimension ℓ , $0 \leq \ell \leq k$.

Theorem

For fixed positive integers k and $n \geq 2k$, the following inequalities hold:

$$EHULL[n, k, 0]_q > EHULL[n, k, 1]_q > \cdots > EHULL[n, k, k]_q. \quad (4)$$

Proven asymptotically by Sendrier! We prove this theorem for all values of n , k and q !

Example: $EHULL[n, k, 0]_2 = 640$, $EHULL[n, k, 1]_2 = 620$,
 $EHULL[n, k, 2]_2 = 120$, $EHULL[n, k, 3]_2 = 15$.

The average dimension of the hull (our research)

- S. Li, M. Shi, Characterization and classification of binary linear codes with various hull dimensions from an improved mass formula, IEEE Trans. Inf. Theory 70(5), 3357–3372 (2024).
- S. Li, M. Shi, S. Ling, A mass formula for linear codes with prescribed hull dimension and related classification, IEEE Trans. Inf. Theory 71(1), 273–286 (2025).

Theorem

Assume that q is even. Let ℓ , k and n be three positive integers such that $\ell \leq k \leq n - \ell$. Suppose that $k_0 = k - \ell$. Then

$$EHULL[n, k, \ell]_q = \begin{cases} \left(\prod_{i=1}^{\ell} \frac{q^{n-k_0-i} - q^{i-1}}{(q^i - 1)q^{k_0+i-1}} \right) q^{(nk_0 - k_0^2 + n - 1)/2} \begin{bmatrix} n/2 - 1 \\ (k_0 - 1)/2 \end{bmatrix}_{q^2} & \begin{array}{l} \text{if } n \text{ is even,} \\ k_0 \text{ is odd,} \end{array} \\ \left(\prod_{i=1}^{\ell} \frac{q^{n-k_0-i} - q^{i-1}}{(q^i - 1)q^{k_0+i-1}} \right) q^{k_0(n-k_0+1)/2} \begin{bmatrix} (n-1)/2 \\ k_0/2 \end{bmatrix}_{q^2} & \begin{array}{l} \text{if } n \text{ is odd,} \\ k_0 \text{ is even,} \end{array} \\ \vdots & \vdots \end{cases} \quad (5)$$

The average dimension of the hull (our research)

Theorem

Let $\ell + 1 \leq k \leq n - \ell - 1$. Then

$$EHULL[n, k, \ell]_q > (q^{\ell+1} - 1)EHULL[n, k, \ell + 1]_q$$

for all possible values of q , n , k and ℓ except when q is odd, $n \equiv 2 \pmod{4}$, -1 is not a square in $\mathbb{F}_q^*$, and $k - \ell$ is odd, in which case $EHULL[n, k, \ell]_q \geq \frac{q^{\ell+1}-1}{2} EHULL[n, k, \ell + 1]_q$.

Conjecture

Let $B_{n,k,\ell,q}$ be the number of all inequivalent linear codes of length n , dimension k and hull dimension ℓ over $\mathbb{F}_q$. If $q = 2$ or 3 and $n \geq 2k$, then

$$\min\{B_{n,k,0,q}, B_{n,k,1,q}\} > B_{n,k,2,q} > \cdots > B_{n,k,k,q}.$$

Experimental results - inequivalent codes

Binary inequivalent $[n, k, \geq 2]$ codes without zero columns:

$k = 6$					
$n =$	12	13	16	17	18
linear	4393	18621	1350439	5548052	22501337
even	303	945	37017	131233	477215
SO	3	3	105	123	521
LCD	1418	5632	484648	2034711	8633817

Ternary inequivalent $[n, k, \geq 2]$ codes without zero columns:

$k = 6$								
$n =$	9	10	11	12	13	14	15	16
linear	137	1638	24342	474106	10956955	?	?	?
SO	-	-	-	3	4	15	61	286
LCD	67	839	13537	283650	?	?	?	?

Experimental results - optimal codes

Stefka Bouyuklieva and Mariya Dzhumalieva-Stoeva. Optimal Linear Codes and their Hulls. Mathematics, 13, 2491, 2025.

Optimal binary codes of dimension $k = 6$

n	d	all codes	h(C)						
			0	1	2	3	4	5	6
17	7	3						3	
18	8	2							2
19	8	28	2		6		11	7	2
20	8	1833	392	85	922	130	255	26	23
21	8	146946	48671	35180	52522	6370	3874	272	57
22	9	248	105	92	35	16			
23	10	29	14		13		2		
24	10	11085	4434	265	6009	57	319		1
25	11	5						5	
26	12	2							2
27	12	178	24		49		63	18	24
28	12	79302	21136	1214	45783	1912	8196	678	383
29	13	2				2			
30	14	2					2		
31	15	1						1	
32	16	1							1

S. Li, M. Shi, Y. Li, S. Ling, A further study on the mass formula for linear codes with prescribed hull dimension, arXiv:2410.13578, 2024.

Theorem

Let n, k, ℓ be positive integers such that $\ell \leq k \leq n - \ell$, and $n_0 = n - k$. Then

$$HHull[n, k, \ell]_{q^2} = \left(\prod_{i=1}^{\ell} M_{n,k,\ell,i} \right) q^{(k-\ell)(n_0+\ell)} \prod_{i=1}^{k-\ell} \frac{q^{n_0+\ell+i} - (-1)^{n_0+\ell+i}}{q^i - (-1)^i},$$

where

$$M(n, k, \ell, i) = \frac{(q^{n_0+\ell-2i+2} - (-1)^{n_0+\ell})(q^{n_0+\ell-2i+1} + (-1)^{n_0+\ell})}{q^{2k-2\ell}(q^{2i} - 1)}.$$

Theorem

Let $1 \leq \ell + 1 \leq k \leq n - k$. Then

$$HHull[n, k, \ell]_{q^2} > \begin{cases} (q^{\ell+1} - 1)HHull[n, k, \ell + 1]_{q^2} & \text{if } k - \ell \text{ is odd,} \\ q^{2\ell}(q - 1)HHull[n, k, \ell + 1]_{q^2} & \text{if } k - \ell \text{ is even.} \end{cases}$$

If there is a linear $[2n, k]_q$ code with ℓ -dimensional symplectic hull, then $k - \ell$ is even.

Theorem

Assume that n , k and ℓ are three nonnegative integers with $\ell \leq k \leq n$ and $k = 2k_0 + \ell$. If $\ell \geq 1$ then

$$SHull[2n, k, \ell]_q = q^{2k_0(n-k_0-\ell)} \begin{bmatrix} n \\ k_0 \end{bmatrix}_{q^2} \prod_{i=1}^{\ell} \frac{q^{2n-2k_0-2i+2} - 1}{q^i - 1}.$$

Furthermore, if k is even, then

$$SHull[2n, k, 0]_q = q^{k(n-k/2)} \begin{bmatrix} n \\ k/2 \end{bmatrix}_{q^2}.$$

Theorem

Assume that n , k and ℓ are three nonnegative integers with $0 \leq \ell \leq k - 2$, $1 \leq k \leq n$, and $k - \ell$ is even. Then

$$SHull[2n, k, \ell]_q > \frac{(q^{\ell+1} - 1)(q^{\ell+2} - 1)}{q^2} SHull[2n, k, \ell + 2]_q.$$

Example

Let $q = 3$, $n = 5$ and $k = 3$. Then

$$SHULL[10, 3, 1]_3 = 17,648,856,720,$$

$$SHULL[10, 3, 3]_3 = 677,871,040, EHULL[10, 3, 3]_2 = 22,408,960.$$

Corollary

If q is even then $SHULL[2n, n, n]_q = (q^n + 1)EHULL[2n, n, n]_q$.

- Comparison between the numbers $EHULL[n, k, \ell]_q$, $HHULL[n, k, \ell]_q$ and $SHULL[n, k, \ell]_q$.
- Symplectic entanglement-assisted stabilizer codes.
- Entanglement-assisted quantum error-correcting codes.

Keita Ishizuka, Comparative monotonicity of linear codes by Hermitian and symplectic hull dimensions, arXiv preprint arXiv:2605.29204, 28 May 2026.

S. Bouyuklieva and I. Bouyukliev are supported by Bulgarian Science Fund under Contract KP-06-H92/3 - 08.12.2025.

Ferruh Özbudak is supported by TÜBİTAK project 223N065.

Thank you for your attention!

Questions?